Neelkumar K. Shah

Cybersecurity Engineer | Blockchain Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Engineer with 5 years of experience securing and building blockchain and web systems. Contributed in the MeitY-funded National Blockchain Framework (NBF) for India's National Blockchain Service, architecting a multi-organization Hyperledger Fabric network and leading vulnerability assessment and penetration testing (VAPT) across 4-5 blockchain applications and 30-45 APIs. Built full-stack security tooling with React, Node.js, Express.js, and MongoDB, including a security assessment platform and a Python database vulnerability scanner that detects 18+ vulnerability classes. Author of 5 peer-reviewed IEEE papers on blockchain and smart contract security.

SKILLS

Cybersecurity: Vulnerability Assessment and Penetration Testing (VAPT), Web Application Security, API Security Testing, Smart Contract Auditing, Threat Modeling, Static Source Code Analysis, NoSQL Database Security Audit, HTTP Security Headers, Content Security Policy (CSP), Security Assessment Automation, Vulnerability Remediation Tracking

Blockchain: Hyperledger Fabric, Hyperledger Sawtooth, Ethereum, Solidity, Node.js Chaincode, Smart Contracts, Decentralized Applications (dApps), Truffle, Ganache, IPFS, MetaMask

Full-Stack Development: JavaScript, Node.js, Express.js, NestJS, React.js, Python, MongoDB, RESTful APIs, Authentication Workflows, HTML/CSS, Bootstrap

Other: Technical Documentation, Security Reporting (PDF), Team Leadership, Training and Curriculum Development

PROFESSIONAL EXPERIENCE

Self-Employed/Freelance

Independent Full-Stack, Blockchain & Security Engineer | Jun 2026 - Present

- Architected and delivered, production-grade corporate air travel web platform for Travel Beyond tbgind.com - an executive air travel management company with operational hubs in Dubai and Mumbai.
- Hardened the Travel Beyond platform with strict Content Security Policy (CSP), and integrated structured schema.org JSON-LD for global SEO.
- Developed Solidity smart contracts for Ethereum-based decentralized applications and performed source code security analysis of Solidity contracts to identify vulnerabilities, logic flaws, and compliance gaps.
- Architected and built Sudarshana, a full-stack web application security assessment platform (Node.js, Express.js, React.js, MongoDB) that automates security assessments of web applications and REST APIs.
- Built RESTful API endpoints, secure authentication workflows, and interactive React.js dashboards with automated PDF audit reports for real-time security posture assessment, vulnerability tracking, and remediation management.

C-DAC: Centre for Development of Advanced Computing

Project Engineer | Pune, Maharashtra, India | Oct 2021 - Jun 2026

- Contributed to the technical build of the National Blockchain Framework (NBF), a MeitY-funded initiative powering India’s National Blockchain Service.

- Designed and architected a unified, multi-organization Hyperledger Fabric network (3 peer nodes, 2 organizations, 1 channel).
- Built REST APIs that integrate blockchain services into security audit workflows.
- Co-led a 7-person team building a full-stack Blockchain Security Assessment Portal (React, Node.js, Express.js, MongoDB) that automates security evaluations across 10-15 APIs, applications, containers, and databases.
- Designed and built NBFDBAudit, a Python-based scanner that detects 18+ vulnerability classes across 4-5 MongoDB/CouchDB environments, including 1 production environment.
- Co-led VAPT across 4-5 blockchain applications and 30-45 APIs, identifying approximately 25 vulnerabilities per application on average, and partnered with engineering teams on remediation.
- Authored and presented 5 peer-reviewed papers at IEEE ICBDS and I2CT conferences on blockchain threat modeling, smart contract vulnerability detection, and NoSQL database security.
- Earned recognition as Performer of the Month at C-DAC.

Huptech Web Private Limited
Jr. UI Designer and Developer | Ahmedabad, Gujarat, India | Jun 2019 - Sep 2019

- Built responsive front-end interfaces using HTML, CSS, JavaScript, jQuery, and Bootstrap.

PROJECTS

Development and Deployment of E-waste Management Modules Using Blockchain Technology
Graduate School of Engineering and Technology, GTU (M.E. Dissertation) | Jun 2020 - Jul 2021

- Designed and built a decentralized Ethereum DApp using Solidity, Truffle, Ganache, and IPFS. that provides tamper-resistant ownership tracking across the e-waste lifecycle.
- Integrated a React.js frontend with MetaMask wallet connectivity for on-chain interactions.
- Conducted threat modeling and optimized the decentralized architecture for performance.

PUBLICATIONS

- A Threat Modeling Approach for Securing Blockchain Framework - 2025 4th IEEE ICBDS
- Security Assessment Framework and Evaluation for Blockchain Applications (SAFE-Block) - 2024 3rd IEEE ICBDS
- Efficient Solution for NoSQL Database Security in Blockchain-Based Applications - 2023 2nd IEEE ICBDS
- Smart Contract Vulnerability Detection Techniques for Hyperledger Fabric - 2023 8th IEEE I2CT
- Smart Contract - Security Assessment Integrated Framework (SC-SIF) for Hyperledger Fabric - 2022 7th IEEE I2CT

EDUCATION

Master's Degree in Cybersecurity

Graduate School of Engineering and Technology, GTU, Ahmedabad, Gujarat, India | Sep 2019 - Jul 2021

Bachelor's Degree in Computer Engineering

Marwadi Education Foundation, Rajkot, Gujarat, India | Aug 2015 - May 2019

CERTIFICATIONS

- Certified Blockchain Security Professional - Blockchain Council, USA
- Blockchain Technology for Government Officials - Future Skills PRIME, NIELIT, Kolkata, India